



AQEEL EBRAHIM
DATA SCIENCE
DSB-1

Buying Raw, Selling Refined

Bahrain's Non-Oil Foreign Trade — An Exploratory Data Analysis

Exploring 704,000+ official trade transactions to map the non-oil diversification story



The Context & Challenge

Deconstructing the headline trade metrics to expose the underlying non-oil structural trade balance.

The Strategic Imperative

The Core Structural Problem

While Bahrain reports a healthy overall headline trade surplus, the reality is that the surplus is almost entirely fueled by hydrocarbons.

Once oil is extracted from the figures, the remaining non-oil trade swings into a dramatic **~1,190 million BD trade deficit** in 2024, emphasizing the critical importance of diversification.

Analytical Scope & Objective

To support Bahrain's Economic Vision 2030, this exploratory data analysis focuses specifically on isolating and assessing the performance of the non-oil economy. By cleaning, mapping, and analyzing transaction-level data, we determine what products the nation imports, what it refines and exports, and where concentration risks lie.

The Data at a Glance



704K+ Transactions

Comprehensive transaction entries bridging annual summaries (2010–2023) and absolute granular records (2023–2024).



6.5K+ HS Codes

High-resolution tracking via standard Harmonized System (HS) lines to map raw commodity categories to finished products.



223 Partners

Deep geographical visibility mapping imports and exports across 223 sovereign global trading partners.

Key Framework Definitions

-  Non-Oil Trade: Represents all commercial international trade transactions, deliberately stripping out petroleum, crude oil, and hydrocarbon derivatives to evaluate the organic diversified sector.
-  National-Origin Export: Industrial goods that are either completely produced, heavily manufactured, or physically refined inside the boundaries of the Kingdom of Bahrain.
-  Re-Export: Foreign-made products imported legally into Bahrain, and subsequently exported to other markets with little to no structural alteration.
-  Trade Balance: Calculated strictly as the sum value of exports minus imports. A negative balance signifies a trade deficit, while a positive value represents a trade surplus.

Trade Flow Dynamics

Isolating and mapping the true magnitude of imports, exports, and geographical partnerships.

The Non-Oil Trade Deficit

1,190M
MILLION BD DEFICIT (2024)

The Hidden Diversification Gap

In 2024, Bahrain's non-oil imports reached approximately **5,872 million BD**, far outstripping non-oil exports which sat at **4,682 million BD**.

Crucially, **~83% of non-oil exports are national-origin products**. This is highly encouraging, indicating that Bahrain's export volume is driven by real, physical domestic value-add rather than superficial re-export trade.

Geographical Concentration



Top Export Destinations

Bahrain's outbound non-oil flows are heavily dependent on adjacent regional partners. **Saudi Arabia and the UAE dominate the export destination portfolio.**

In total, the top five export destination countries account for a significant **54% of Bahrain's entire non-oil export value**, introducing geographical risk.



Top Import Sources

By contrast, Bahrain's import network is geographically diverse. **China, Australia, and Brazil** represent the primary import partners.

The top five import sources account for a combined **46% of total imports**. These sources represent the core pipeline for raw metallurgical raw materials.

Industrial Value Chain

Value-Added Refinement

The exploratory data analysis reveals a clear economic cycle:
Bahrain acts as an industrial refiner.

The top import lines are unrefined materials—chiefly **iron ores and alumina**. Utilizing competitive local energy supplies, local manufacturers refine these raw inputs.

They are subsequently exported as high-value commodities:
unwrought aluminium alloys, structural steel segments, and advanced chemical products.



Strategic Policy Directives



1. Import Substitution

Evaluate local production of major high-value import categories (like electronics and specialized vehicles) to structurally reduce the non-oil deficit.



2. Downstream Integration

Deepen the downstream value chain for aluminium and steel products. Exporting finished assemblies captures higher margin per ton than unwrought alloy bars.



3. Export Diversification

Build trade bridges beyond Saudi Arabia and the UAE. Actively target emerging markets to decouple the country from localized macroeconomic shocks.

Discussion & Next Steps

Where does the research go from here?

Panel Extension

Integrate complete multi-year transaction trends spanning 2020 through 2026.

Macro Reintegration

Layer oil trade flows back in to build a fully reconciled overview of total trade.

Inflation Adjustments

Re-estimate trade values adjusting for global commodities index inflation.

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Image Sources



<https://www.legalwiz.in/blog/how-to-start-import-export-business-in-india>



<https://www.focoinduction.com/wp-content/uploads/2025/08/aluminum-melting.jpg.webp>

Source: www.focoinduction.com